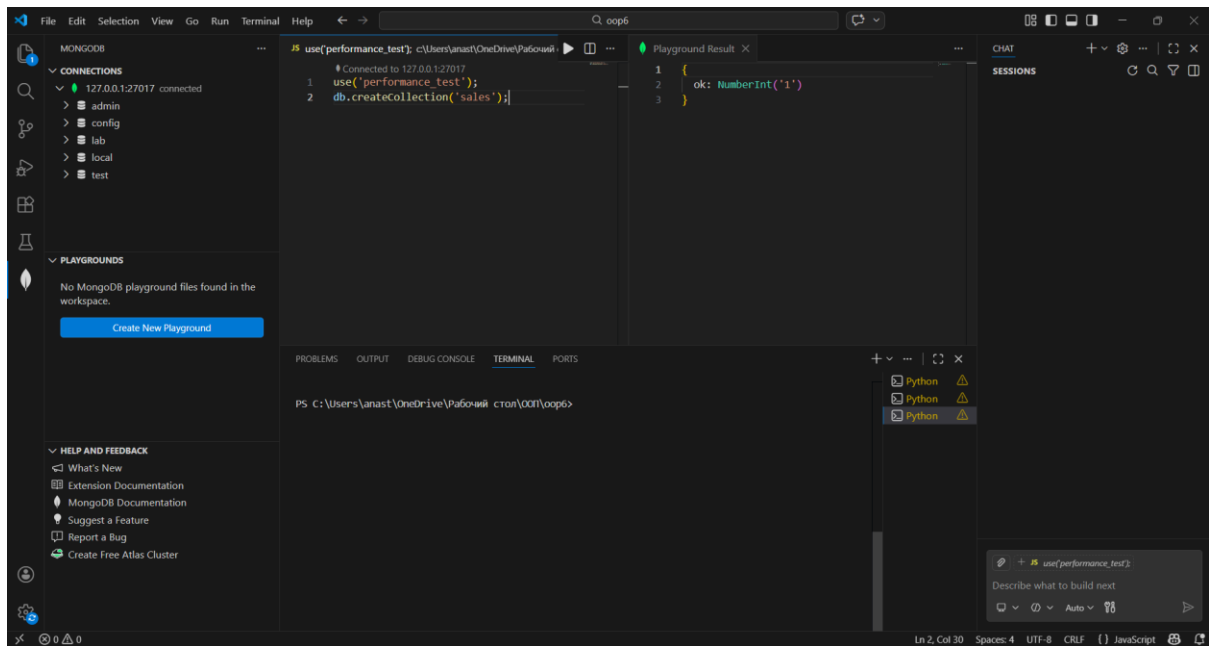


## Лабораторна робота № 6

### Сластнікової Анастасії

**Тема:** Поглиблене вивчення MongoDB: оптимізація продуктивності, використання шардінгу та реплікації

**Мета:** Ознайомитися з методами підвищення продуктивності MongoDB. Навчитися використовувати індекси для оптимізації запитів. Дослідити механізми реплікації та шардінгу.



Створення бази даних та колекції в середовищі VS Code

```

main.py > ...
1  port pymongo
2  port random
3  from datetime import datetime, timedelta
4
5  1. Підключення до локального сервера MongoDB
6  client = pymongo.MongoClient("mongodb://localhost:27017/")
7
8  2. Вибір бази даних та колекції
9  db = client["performance_test"]
10 collection = db["sales"]
11
12 Очищення колекції перед вставкою, щоб уникнути дублювання при повтор
13 collection.delete_many({})
14
15 Списки для генерації випадкових даних
16 categories = ["Electronics", "Clothing", "Home", "Books", "Toys"]
17 data_to_insert = []
18 total_documents = 100000
19 batch_size = 5000 # Вставка порціями для кращої продуктивності
20
21 print(f"Початок генерації {total_documents} документів...")
22
23 for i in range(total_documents):
24     # Створення одного документа
25     document = {
26         "orderId": f"ORD{i:06d}",
27         "category": random.choice(categories),
28         "price": round(random.uniform(10.0, 1500.0), 2),
29         "quantity": random.randint(1, 10),
30         "timestamp": datetime.now() - timedelta(days=random.randint(0,
31     })
32     data_to_insert.append(document)
33
34 # Якщо накопичилася порція (batch), вставляємо її в базу
35 if len(data_to_insert) == batch_size:
36     collection.insert_many(data_to_insert)
37     data_to_insert = []

```

```

33
34 Click to add a breakpoint
35 if len(data_to_insert) == batch_size:
36     collection.insert_many(data_to_insert)
37     data_to_insert = []
38     print(f"Вставлено {i + 1} документів...")
39
40 Вставка залишку, якщо він є
41 data_to_insert:
42     collection.insert_many(data_to_insert)
43
44 print("Генерація та вставка завершена успішно!")
45 print(f"Загальна кількість документів у колекції 'sales': {collection.

```

Реалізація скрипту Python для генерації тестових даних

```
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6> & C:/Users/anast/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/anast/OneDrive/Робочий стол/бд/lab6/main.py"
Початок генерації 100000 документів...
Вставлено 5000 документів...
Вставлено 10000 документів...
Вставлено 15000 документів...
Вставлено 20000 документів...
Вставлено 25000 документів...
Вставлено 30000 документів...
Вставлено 35000 документів...
Вставлено 40000 документів...
Вставлено 45000 документів...
Вставлено 50000 документів...
Вставлено 55000 документів...
Вставлено 60000 документів...
Вставлено 65000 документів...
Вставлено 70000 документів...
Вставлено 75000 документів...
Вставлено 80000 документів...
Вставлено 85000 документів...
Вставлено 90000 документів...
Вставлено 95000 документів...
Вставлено 100000 документів...
Генерація та вставка завершена успішно!
Загальна кількість документів у колекції 'sales': 100000
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6>
```

Підтвердження початку генерації 100 000 документів.

The screenshot shows the Visual Studio Code editor with the file explorer on the left displaying a project named 'LAB6' containing 'main.py' and 'query\_no\_index.py'. The 'query\_no\_index.py' file is open in the editor, showing a Python script that connects to a MongoDB database, finds documents in the 'sales' collection where the category is 'Electronics', and prints the results. The terminal at the bottom shows the output of the script, indicating that 28025 documents were found and the execution time was 0.4954 seconds.

```
query_no_index.py > ...
1  t = pymongo
2  t = time
3
4  t = pymongo.MongoClient("mongodb://localhost:27017/")
5  ction = client["performance_test"]["sales"]
6
7  ірюємо час початку
8  _time = time.time()
9
10 онуємо пошук за категорією
11 ts = list(collection.find({"category": "Electronics"}))
12
13 ірюємо час завершення
14 ime = time.time()
15
16 (f"Знайдено документів: {len(results)}")
17 [(f"Час виконання БЕЗ індексу: {end_time - start_time:.4f} секунд")]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

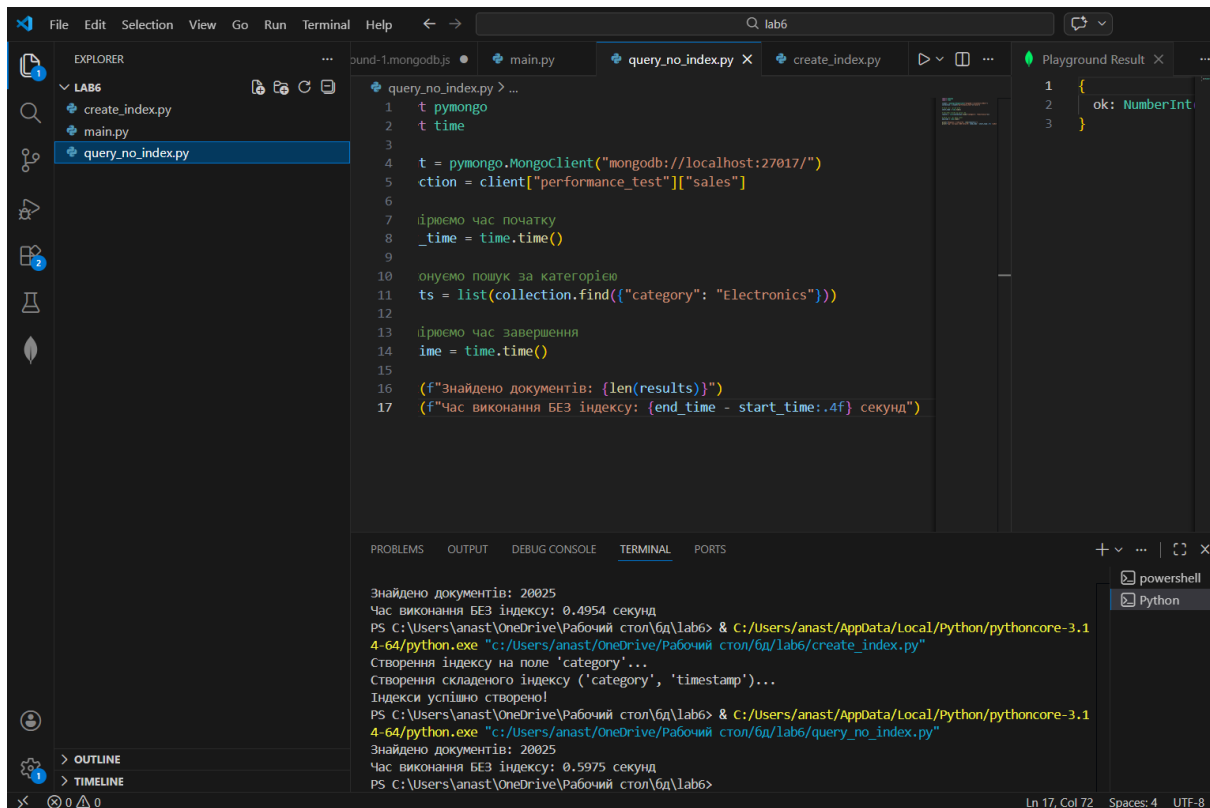
Загальна кількість документів у колекції 'sales': 100000  
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6> & c:/Users/anast/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/anast/OneDrive/Робочий стол/бд/lab6/query\_no\_index.py"  
Знайдено документів: 28025  
Час виконання БЕЗ індексу: 0.4954 секунд  
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6>

The screenshot shows the Visual Studio Code editor with the file explorer on the left displaying a project named 'LAB6' containing 'main.py', 'query\_no\_index.py', and 'create\_index.py'. The 'create\_index.py' file is open in the editor, showing a Python script that connects to a MongoDB database and creates two indexes on the 'sales' collection: a simple index on the 'category' field and a compound index on ('category', 'timestamp'). The terminal at the bottom shows the output of the script, indicating that the indexes were successfully created.

```
create_index.py > ...
1  import pymongo
2
3  client = pymongo.MongoClient("mongodb://localhost:27017/")
4  collection = client["performance_test"]["sales"]
5
6  # 1. Створення простого індексу
7  print("Створення індексу на поле 'category'...")
8  collection.create_index([("category", 1)])
9
10 # 2. Створення складеного індексу
11 print("Створення складеного індексу ('category', 'timestamp')...")
12 collection.create_index([("category", 1), ("timestamp", -1)])
13
14 print(f"Індекси успішно створено!")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

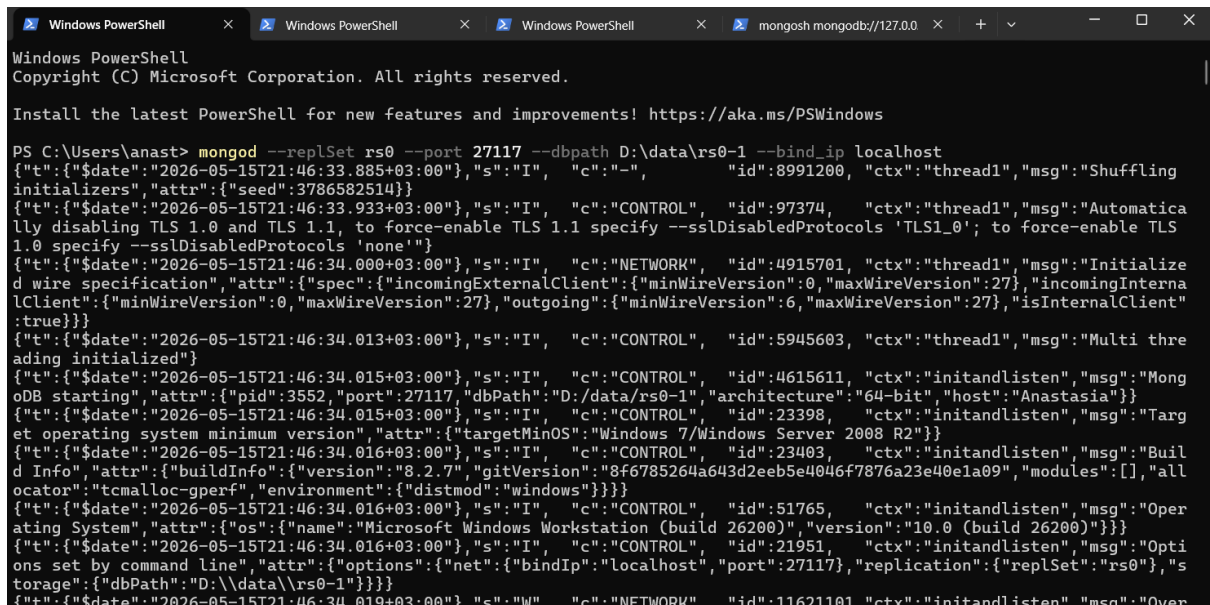
4-64/python.exe "c:/Users/anast/OneDrive/Робочий стол/бд/lab6/query\_no\_index.py"  
Знайдено документів: 28025  
Час виконання БЕЗ індексу: 0.4954 секунд  
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6> & c:/Users/anast/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/anast/OneDrive/Робочий стол/бд/lab6/create\_index.py"  
Створення індексу на поле 'category'...  
Створення складеного індексу ('category', 'timestamp')...  
Індекси успішно створено!  
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6>



```
query_no_index.py > ...
1 t pymongo
2 t time
3
4 t = pymongo.MongoClient("mongodb://localhost:27017/")
5 ction = client["performance_test"]["sales"]
6
7 ірємо час початку
8 _time = time.time()
9
10 онуємо пошук за категорією
11 ts = list(collection.find({"category": "Electronics"}))
12
13 ірємо час завершення
14 ime = time.time()
15
16 (f"Знайдено документів: {len(results)}")
17 (f"Час виконання БЕЗ індексу: {end_time - start_time:.4f} секунд")
```

```
Знайдено документів: 20025
Час виконання БЕЗ індексу: 0.4954 секунд
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6> & C:/Users/anast/AppData/Local/Python/pythoncore-3.1
4-64/python.exe "c:/Users/anast/OneDrive/Робочий стол/бд/lab6/create_index.py"
Створення індексу на поле 'category'...
Створення складеного індексу ('category', 'timestamp')...
Індекси успішно створено!
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6> & C:/Users/anast/AppData/Local/Python/pythoncore-3.1
4-64/python.exe "c:/Users/anast/OneDrive/Робочий стол/бд/lab6/query_no_index.py"
Знайдено документів: 20025
Час виконання БЕЗ індексу: 0.5975 секунд
PS C:\Users\anast\OneDrive\Робочий стол\бд\lab6>
```

## Вимірювання часу виконання запиту без використання індексів



```
Windows PowerShell
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PS C:\Users\anast> mongod --replSet rs0 --port 27117 --dbpath D:\data\rs0-1 --bind_ip localhost
{"t":{"$date":"2026-05-15T21:46:33.885+03:00"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1", "msg":"Shuffling
initializers", "attr":{"seed":3786582514}}
{"t":{"$date":"2026-05-15T21:46:33.933+03:00"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1", "msg":"Automatica
lly disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS
1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-05-15T21:46:34.000+03:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1", "msg":"Initialize
d wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingIntern
alClient":{"minWireVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient"
:true}}}}
{"t":{"$date":"2026-05-15T21:46:34.013+03:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1", "msg":"Multi thre
ading initialized"}
{"t":{"$date":"2026-05-15T21:46:34.015+03:00"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten", "msg":"Mong
oDB starting", "attr":{"pid":3552, "port":27117, "dbPath":"D:/data/rs0-1", "architecture":"64-bit", "host":"Anastasia"}}
{"t":{"$date":"2026-05-15T21:46:34.015+03:00"},"s":"I", "c":"CONTROL", "id":23398, "ctx":"initandlisten", "msg":"Targ
et operating system minimum version", "attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"$date":"2026-05-15T21:46:34.016+03:00"},"s":"I", "c":"CONTROL", "id":23403, "ctx":"initandlisten", "msg":"Buil
d Info", "attr":{"buildInfo":{"version":"8.2.7", "gitVersion":"8f6785264a643d2eeb5e4046f7876a23e40e1a09", "modules":[],"all
ocator":"tcmalloc-gperf", "environment":{"distmod":"windows"}}}}
{"t":{"$date":"2026-05-15T21:46:34.016+03:00"},"s":"I", "c":"CONTROL", "id":51765, "ctx":"initandlisten", "msg":"Oper
ating System", "attr":{"os":{"name":"Microsoft Windows Workstation (build 26200)", "version":"10.0 (build 26200)}}}}
{"t":{"$date":"2026-05-15T21:46:34.016+03:00"},"s":"I", "c":"CONTROL", "id":21951, "ctx":"initandlisten", "msg":"Opti
ons set by command line", "attr":{"options":{"net":{"bindIp":"localhost", "port":27117, "replication":{"replSet":"rs0"},"s
torage":{"dbPath":"D:\\data\\rs0-1"}}}}}}
{"t":{"$date":"2026-05-15T21:46:34.019+03:00"},"s":"W", "c":"NETWORK", "id":11621101, "ctx":"initandlisten", "msg":"Over
```

## Запуск экземпляра базы данных на порту 27117

```
Windows PowerShell
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PS C:\Users\anast> mongod --replSet rs0 --port 27018 --dbpath D:\data\rs0-2 --bind_ip localhost
{"t":{"$date":"2026-05-15T21:46:47.517+03:00"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1", "msg":"Shuffling
initializers", "attr":{"seed":2085433181}}
{"t":{"$date":"2026-05-15T21:46:47.572+03:00"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1", "msg":"Automatica
lly disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS
1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-05-15T21:46:47.576+03:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1", "msg":"Initialize
d wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingIntern
alClient":{"minWireVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient"
:true}}}
{"t":{"$date":"2026-05-15T21:46:47.577+03:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1", "msg":"Multi thre
ading initialized"}
{"t":{"$date":"2026-05-15T21:46:47.577+03:00"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten", "msg":"Mong
oDB starting", "attr":{"pid":20036, "port":27018, "dbPath":"D:/data/rs0-2", "architecture":"64-bit", "host":"Anastasia"}}
{"t":{"$date":"2026-05-15T21:46:47.577+03:00"},"s":"I", "c":"CONTROL", "id":23398, "ctx":"initandlisten", "msg":"Targ
et operating system minimum version", "attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"$date":"2026-05-15T21:46:47.577+03:00"},"s":"I", "c":"CONTROL", "id":23403, "ctx":"initandlisten", "msg":"Buil
d Info", "attr":{"buildInfo":{"version":"8.2.7", "gitVersion":"8f6785264a643d2eeb5e4046f7876a23e40e1a09", "modules":[], "all
ocator":"tcmalloc-gperf", "environment":{"distmod":"windows"}}}}
{"t":{"$date":"2026-05-15T21:46:47.577+03:00"},"s":"I", "c":"CONTROL", "id":51765, "ctx":"initandlisten", "msg":"Oper
ating System", "attr":{"os":{"name":"Microsoft Windows Workstation (build 26200)", "version":"10.0 (build 26200)"}}}
{"t":{"$date":"2026-05-15T21:46:47.577+03:00"},"s":"I", "c":"CONTROL", "id":21951, "ctx":"initandlisten", "msg":"Opti
ons set by command line", "attr":{"options":{"net":{"bindIp":"localhost", "port":27018}, "replication":{"replSet":"rs0"},"s
torage":{"dbPath":"D:\\data\\rs0-2"}}}}
{"t":{"$date":"2026-05-15T21:46:47.577+03:00"},"s":"W", "c":"NETWORK", "id":11621101, "ctx":"initandlisten", "msg":"Over
```

Запуск экземпляра базы даних на порту 27018

```
Windows PowerShell
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PS C:\Users\anast> mongod --replSet rs0 --port 27019 --dbpath D:\data\rs0-3 --bind_ip localhost
{"t":{"$date":"2026-05-15T21:46:55.695+03:00"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1", "msg":"Shuffling
initializers", "attr":{"seed":3242535344}}
{"t":{"$date":"2026-05-15T21:46:55.763+03:00"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1", "msg":"Automatica
lly disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS
1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-05-15T21:46:55.769+03:00"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1", "msg":"Initialize
d wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingIntern
alClient":{"minWireVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient"
:true}}}
{"t":{"$date":"2026-05-15T21:46:55.769+03:00"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1", "msg":"Multi thre
ading initialized"}
{"t":{"$date":"2026-05-15T21:46:55.769+03:00"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten", "msg":"Mong
oDB starting", "attr":{"pid":10204, "port":27019, "dbPath":"D:/data/rs0-3", "architecture":"64-bit", "host":"Anastasia"}}
{"t":{"$date":"2026-05-15T21:46:55.769+03:00"},"s":"I", "c":"CONTROL", "id":23398, "ctx":"initandlisten", "msg":"Targ
et operating system minimum version", "attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"$date":"2026-05-15T21:46:55.769+03:00"},"s":"I", "c":"CONTROL", "id":23403, "ctx":"initandlisten", "msg":"Buil
d Info", "attr":{"buildInfo":{"version":"8.2.7", "gitVersion":"8f6785264a643d2eeb5e4046f7876a23e40e1a09", "modules":[], "all
ocator":"tcmalloc-gperf", "environment":{"distmod":"windows"}}}}
{"t":{"$date":"2026-05-15T21:46:55.769+03:00"},"s":"I", "c":"CONTROL", "id":51765, "ctx":"initandlisten", "msg":"Oper
ating System", "attr":{"os":{"name":"Microsoft Windows Workstation (build 26200)", "version":"10.0 (build 26200)"}}}
{"t":{"$date":"2026-05-15T21:46:55.769+03:00"},"s":"I", "c":"CONTROL", "id":21951, "ctx":"initandlisten", "msg":"Opti
ons set by command line", "attr":{"options":{"net":{"bindIp":"localhost", "port":27019}, "replication":{"replSet":"rs0"},"s
torage":{"dbPath":"D:\\data\\rs0-3"}}}}
{"t":{"$date":"2026-05-15T21:46:55.770+03:00"},"s":"W", "c":"NETWORK", "id":11621101, "ctx":"initandlisten", "msg":"Over
```

Запуск экземпляра базы даних на порту 27019

```
Windows PowerShell
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PS C:\Users\anast> mongosh --port 27117
Current Mongosh Log ID: 6a076a2777f753f4b9abc113
Connecting to:  mongodb://127.0.0.1:27117/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB: 8.2.7
Using Mongosh: 2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2026-05-15T21:46:34.126+03:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> rs.initiate()
{
  info2: 'no configuration specified. Using a default configuration for the set',
  me: 'localhost:27117',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778870971, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778870971, i: 1 })
}
```

## Аналіз технічного статусу та топології Replica Set

```
}
},
operationTime: Timestamp({ t: 1778870971, i: 1 })
}
rs0 [direct: secondary] test> rs.add("localhost:27018")
|
{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778870981, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778870981, i: 1 })
}
rs0 [direct: primary] test> rs.add("localhost:27019")
|
{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778870992, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778870992, i: 1 })
}
```

## Додавання вторинних вузлів до кластера

```

    },
    operationTime: Timestamp({ t: 1778870992, i: 1 })
  }
}
rs0 [direct: primary] test> rs.status()
{
  set: 'rs0',
  date: ISODate('2026-05-15T18:50:03.673Z'),
  myState: 1,
  term: Long('1'),
  syncSourceHost: '',
  syncSourceId: -1,
  heartbeatIntervalMillis: Long('2000'),
  majorityVoteCount: 2,
  writeMajorityCount: 2,
  votingMembersCount: 3,
  writableVotingMembersCount: 3,
  optimes: {
    lastCommittedOpTime: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
    lastCommittedWallTime: ISODate('2026-05-15T18:49:54.691Z'),
    readConcernMajorityOpTime: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
    appliedOpTime: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
    durableOpTime: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
    writtenOpTime: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
    lastAppliedWallTime: ISODate('2026-05-15T18:49:54.691Z'),
    lastDurableWallTime: ISODate('2026-05-15T18:49:54.691Z'),
    lastWrittenWallTime: ISODate('2026-05-15T18:49:54.691Z')
  },
  lastStableRecoveryTimestamp: Timestamp({ t: 1778870971, i: 1 }),
  electionCandidateMetrics: {
    lastElectionReason: 'electionTimeout',

```

Перевірка розподілу ролей та мережевої доступності вузлів кластера

```

    lastStableRecoveryTimestamp: Timestamp({ t: 1778870971, i: 1 }),
    electionCandidateMetrics: {
      lastElectionReason: 'electionTimeout',
      lastElectionDate: ISODate('2026-05-15T18:49:31.249Z'),
      electionTerm: Long('1'),
      lastCommittedOpTimeAtElection: { ts: Timestamp({ t: 1778870971, i: 1 }), t: Long('-1') },
      lastSeenWrittenOpTimeAtElection: { ts: Timestamp({ t: 1778870971, i: 1 }), t: Long('-1') },
      lastSeenOpTimeAtElection: { ts: Timestamp({ t: 1778870971, i: 1 }), t: Long('-1') },
      numVotesNeeded: 1,
      priorityAtElection: 1,
      electionTimeoutMillis: Long('10000'),
      newTermStartDate: ISODate('2026-05-15T18:49:31.287Z'),
      wMajorityWriteAvailabilityDate: ISODate('2026-05-15T18:49:31.365Z')
    },
    members: [
      {
        _id: 0,
        name: 'localhost:27117',
        health: 1,
        state: 1,
        stateStr: 'PRIMARY',
        uptime: 210,
        optime: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
        optimeDate: ISODate('2026-05-15T18:49:54.000Z'),
        optimeWritten: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
        optimeWrittenDate: ISODate('2026-05-15T18:49:54.000Z'),
        lastAppliedWallTime: ISODate('2026-05-15T18:49:54.691Z'),
        lastDurableWallTime: ISODate('2026-05-15T18:49:54.691Z'),
        lastWrittenWallTime: ISODate('2026-05-15T18:49:54.691Z'),
        syncSourceHost: ''

```

Аналіз стану головного вузла (Primary) у реплікаційному наборі.



```

syncSourceId: -1,
infoMessage: 'Could not find member to sync from',
electionTime: Timestamp({ t: 1778870971, i: 2 }),
electionDate: ISODate('2026-05-15T18:49:31.000Z'),
configVersion: 5,
configTerm: 1,
self: true,
lastHeartbeatMessage: ''
},
{
  _id: 1,
  name: 'localhost:27018',
  health: 1,
  state: 2,
  stateStr: 'SECONDARY',
  uptime: 22,
  optime: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
  optimeDurable: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
  optimeWritten: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
  optimeDate: ISODate('2026-05-15T18:49:54.000Z'),
  optimeDurableDate: ISODate('2026-05-15T18:49:54.000Z'),
  optimeWrittenDate: ISODate('2026-05-15T18:49:54.000Z'),
  lastAppliedWallTime: ISODate('2026-05-15T18:49:54.691Z'),
  lastDurableWallTime: ISODate('2026-05-15T18:49:54.691Z'),
  lastWrittenWallTime: ISODate('2026-05-15T18:49:54.691Z'),
  lastHeartbeat: ISODate('2026-05-15T18:50:02.744Z'),
  lastHeartbeatRecv: ISODate('2026-05-15T18:50:02.743Z'),
  pingMs: Long('0'),
  lastHeartbeatMessage: '',
  syncSourceHost: 'localhost:27117',

```

Моніторинг стану вторинного вузла (Secondary) та параметрів синхронізації.

```

syncSourceId: 0,
infoMessage: '',
configVersion: 5,
configTerm: 1
},
{
  _id: 2,
  name: 'localhost:27019',
  health: 1,
  state: 2,
  stateStr: 'SECONDARY',
  uptime: 10,
  optime: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
  optimeDurable: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
  optimeWritten: { ts: Timestamp({ t: 1778870994, i: 1 }), t: Long('1') },
  optimeDate: ISODate('2026-05-15T18:49:54.000Z'),
  optimeDurableDate: ISODate('2026-05-15T18:49:54.000Z'),
  optimeWrittenDate: ISODate('2026-05-15T18:49:54.000Z'),
  lastAppliedWallTime: ISODate('2026-05-15T18:49:54.691Z'),
  lastDurableWallTime: ISODate('2026-05-15T18:49:54.691Z'),
  lastWrittenWallTime: ISODate('2026-05-15T18:49:54.691Z'),
  lastHeartbeat: ISODate('2026-05-15T18:50:02.744Z'),
  lastHeartbeatRecv: ISODate('2026-05-15T18:50:02.743Z'),
  pingMs: Long('0'),
  lastHeartbeatMessage: '',
  syncSourceHost: 'localhost:27018',
  syncSourceId: 1,
  infoMessage: '',
  configVersion: 5,
  configTerm: 1

```

Аналіз стану та мережевих зв'язків третього вузла реплікаційного набору.

```

    }
  ],
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778870994, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778870994, i: 1 })
}
rs0 [direct: primary] test> |

```

Фінальне підтвердження працездатності та синхронізації Replica Set.

## New Connection ✕

Manage your connection settings

**URI** ⓘ Edit Connection String ☐

mongodb://localhost:27117,localhost:27018,localhost:27019/?replicaSet=rs0

**Name**

localhost:27117,localhost:27018,localhost:27019

**Color**

No Color

☐ **Favorite this connection**  
 Favoriting a connection will pin it to the top of your list of connections

➤ **Advanced Connection Options**

**How do I find my connection string in Atlas?**

If you have an Atlas cluster, go to the Cluster view. Click the 'Connect' button for the cluster to which you wish to connect.

[See example](#)

**How do I format my connection string?**

[See example](#)

Cancel

Save

Connect

Save & Connect

Налаштування підключення до Replica Set у MongoDB Compass.

MongoDB Compass - localhost:27117,localhost:27018,localhost:27019/local

Connections Edit View Help

Welcome local +

localhost:27117,localhost:27018,localhost:27019 > local

Open MongoDB shell Create collection Refresh

| Collection name                 | Properties | Storage size | Data size | Documents | Avg. document size | Indexes | Total index size |
|---------------------------------|------------|--------------|-----------|-----------|--------------------|---------|------------------|
| oplog.rs                        | CAPPED     | 45.06 kB     | 30.13 kB  | 260       | 115.00 B           | 0       | 0 B              |
| replset.election                | -          | 36.86 kB     | 60.00 B   | 1         | 60.00 B            | 1       | 20.48 kB         |
| replset.initialSyncId           | -          | 20.48 kB     | 49.00 B   | 1         | 49.00 B            | 1       | 20.48 kB         |
| replset.minInvalid              | -          | 20.48 kB     | 45.00 B   | 1         | 45.00 B            | 1       | 20.48 kB         |
| replset.oplogTruncateAfterPoint | -          | 36.86 kB     | 71.00 B   | 1         | 71.00 B            | 1       | 20.48 kB         |
| startup_log                     | CAPPED     | 20.48 kB     | 3.55 kB   | 1         | 3.55 kB            | 1       | 20.48 kB         |

CONNECTIONS (3)

Search connections

- localhost:27017
- localhost:27017
- localhost:27117,localhost:27018,localhost:27019
  - admin
  - config
  - local
    - oplog.rs
    - replset.election
    - replset.initialSyncId
    - replset.minInvalid
    - replset.oplogTruncateAfterPoint
    - startup\_log

Візуалізація системних колекцій бази даних local у MongoDB Compass

```

Windows PowerShell
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PS C:\Users\anast> mkdir D:\data\cfg

    Directory: D:\data

Mode                LastWriteTime         Length Name
----                -
d-----          15.05.2026    22:35             cfg

PS C:\Users\anast> mkdir D:\data\shard1

    Directory: D:\data

Mode                LastWriteTime         Length Name
----                -
d-----          15.05.2026    22:36             shard1

PS C:\Users\anast> mkdir D:\data\shard2

    Directory: D:\data

Mode                LastWriteTime         Length Name
----                -
d-----          15.05.2026    22:36             shard2

PS C:\Users\anast> mongod --configsvr --replSet cfg --port 27030 --dbpath D:\data\cfg --bind_ip localhost
{"t":{"date":"2026-05-15T22:36:55.861+03:00"},"s":"I",  "c":"-",               "id":8991200, "ctx":"thread1","msg":"Shuffling initializers","attr":{"seed":3016729
106}}

```

Підготовка середовища та запуск Config Server для кластера шардінгу.

```

Windows PowerShell
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PS C:\Users\anast> mongosh --port 27030
Current Mongosh Log ID: 6a0775e7e8d8bdb2d2abc113
Connecting to:      mongodb://127.0.0.1:27030/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB:      8.2.7
Using Mongosh:       2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-05-15T22:36:56.060+03:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> rs.initiate({
  _id: "cfg",
  configsvr: true,
  members: [
    { _id: 0, host: "localhost:27030" }
  ]
})
{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778873840, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778873840, i: 1 })
}

```

Ініціалізація набору реплік для Config Server.

```

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\anast> mongod --shardsvr --replSet shard1rs --port 27021 --dbpath D:\data\shard1 --bind_ip localhost
{"t":{"$date":"2026-05-15T23:04:38.570+03:00"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1","msg":"Shuffling initializers","attr":{"seed":3954349291}}
{"t":{"$date":"2026-05-15T23:04:38.615+03:00"},"s":"I", "c":"CONTROL", "id":97374, "svc":"-", "ctx":"thread1","msg":"Automatically disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-05-15T23:04:38.676+03:00"},"s":"I", "c":"NETWORK", "id":4915701, "svc":"-", "ctx":"thread1","msg":"Initialized wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient":true}}}
{"t":{"$date":"2026-05-15T23:04:38.685+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"ShardingDDLCoordinator","namespace":"config.system.sharding_ddl_coordinators"}}
{"t":{"$date":"2026-05-15T23:04:38.685+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"RenameCollectionParticipantService","namespace":"config.localRenameParticipants"}}
{"t":{"$date":"2026-05-15T23:04:38.685+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"ReshardingDonorService","namespace":"config.localReshardingOperations.donor"}}
{"t":{"$date":"2026-05-15T23:04:38.685+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"ReshardingRecipientService","namespace":"config.localReshardingOperations.recipient"}}
{"t":{"$date":"2026-05-15T23:04:38.686+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"MultiUpdateCoordinatorService","namespace":"config.localMigrationBlockingOperations.multiUpdateCoordinators"}}
{"t":{"$date":"2026-05-15T23:04:38.686+03:00"},"s":"I", "c":"CONTROL", "id":5945603, "svc":"-", "ctx":"thread1","msg":"Multi threading initialized"}
{"t":{"$date":"2026-05-15T23:04:38.687+03:00"},"s":"I", "c":"CONTROL", "id":4615611, "svc":"S", "ctx":"initandlisten","msg":"MongoDB starting","attr":{"pid":8620,"port":27021,"dbPath":"D:/data/shard1","architecture":"64-bit","host":"Anastasia"}}
{"t":{"$date":"2026-05-15T23:04:38.687+03:00"},"s":"I", "c":"CONTROL", "id":23398, "svc":"S", "ctx":"initandlisten","msg":"Target operating system minimum version","attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"$date":"2026-05-15T23:04:38.687+03:00"},"s":"I", "c":"CONTROL", "id":23403, "svc":"S", "ctx":"initandlisten","msg":"Build Info","attr":{"buildInfo":{"version":"8.2.7","gitVersion":"8f6785264a643d2eeb5e4046f7876a23e40e1a09","modules":[],"allocator":"tcmalloc-gperf","environment":{"distmod":"windows"}}}}
{"t":{"$date":"2026-05-15T23:04:38.687+03:00"},"s":"I", "c":"CONTROL", "id":51765, "svc":"S", "ctx":"initandlisten","msg":"Operating System","attr":{"os":{"name":"Microsoft Windows Workstation (build 26200)","version":"18.0 (build 26200)}}}}
{"t":{"$date":"2026-05-15T23:04:38.687+03:00"},"s":"I", "c":"CONTROL", "id":21951, "svc":"S", "ctx":"initandlisten","msg":"Options set by command line","attr":{"options":{"net":{"bindIp":"localhost","port":27021},"replication":{"replSet":"shard1rs"},"sharding":{"clusterRole":"shardsvr"},"storage":{"dbPath":"D:/data/shard1"}}}}}
{"t":{"$date":"2026-05-15T23:04:38.688+03:00"},"s":"W", "c":"NETWORK", "id":11621101,"svc":"S", "ctx":"initandlisten","msg":"Overriding max connections to honor capMemoryConsumptionForPreAuthBuffers settings","attr":{"limit":100939}}
{"t":{"$date":"2026-05-15T23:04:38.689+03:00"},"s":"I", "c":"NETWORK", "id":4648601, "svc":"S", "ctx":"initandlisten","msg":"Implicit TCP FastOpen unavailable. If TCP FastOpen is required, set at least one of the related parameters","attr":{"relatedParameters":["tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"]}}}

```

Запуск першого вузла зберігання даних (Shard Server) у системі шардінгу.

```

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\anast> mongosh --port 27021
Current Mongosh Log ID: 6a077c8566f7024b1eabc113
Connecting to:      mongod://127.0.0.1:27021/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB:      8.2.7
Using Mongosh:       2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-05-15T23:04:38.819+03:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
  -----

test> rs.initiate({
  _id: "shard1rs",
  members: [
    { _id: 0, host: "localhost:27021" }
  ]
})

{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778875541, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778875541, i: 1 })
}
shard1rs [direct: secondary] test>

```

Ініціалізація набору реплік для першого шарду кластера.

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\anast> mongod --shardsvr --replSet shard2rs --port 27022 --dbpath D:\data\shard2 --bind_ip localhost
{"t":{"$date":"2026-05-15T23:06:05.048+03:00"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1","msg":"Shuffling initializers","attr":{"seed":862870543}}
{"t":{"$date":"2026-05-15T23:06:05.090+03:00"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1","msg":"Automatically disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0' to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-05-15T23:06:05.154+03:00"},"s":"I", "c":"NETWORK", "id":4915701, "svc":"-", "ctx":"thread1","msg":"Initialized wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient":true}}}
{"t":{"$date":"2026-05-15T23:06:05.156+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"ShardingDDLCoordinator","namespace":"config.system.sharding_ddl_coordinators"}}
{"t":{"$date":"2026-05-15T23:06:05.156+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"RenameCollectionParticipantService","namespace":"config.localRenameParticipants"}}
{"t":{"$date":"2026-05-15T23:06:05.156+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"ReshardingDonorService","namespace":"config.localReshardingOperations.donor"}}
{"t":{"$date":"2026-05-15T23:06:05.156+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"ReshardingRecipientService","namespace":"config.localReshardingOperations.recipient"}}
{"t":{"$date":"2026-05-15T23:06:05.156+03:00"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1","msg":"Successfully registered PrimaryOnly Service","attr":{"service":"MultiUpdateCoordinatorService","namespace":"config.localMigrationBlockingOperations.multiUpdateCoordinators"}}
{"t":{"$date":"2026-05-15T23:06:05.157+03:00"},"s":"I", "c":"CONTROL", "id":5945603, "svc":"-", "ctx":"thread1","msg":"Multi threading initialized"}
{"t":{"$date":"2026-05-15T23:06:05.157+03:00"},"s":"I", "c":"CONTROL", "id":4615611, "svc":"S", "ctx":"initandlisten","msg":"MongoDB starting","attr":{"pid":24016,"port":27022,"dbPath":"D:/data/shard2","architecture":"64-bit","host":"Anastasia"}}
{"t":{"$date":"2026-05-15T23:06:05.157+03:00"},"s":"I", "c":"CONTROL", "id":23398, "svc":"S", "ctx":"initandlisten","msg":"Target operating system minimum version","attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"$date":"2026-05-15T23:06:05.157+03:00"},"s":"I", "c":"CONTROL", "id":23403, "svc":"S", "ctx":"initandlisten","msg":"Build Info","attr":{"buildInfo":{"version":"8.2.7","gitVersion":"8f6785264a643d2eeb5e4046f7876a23e40e1a09","modules":[],"allocator":"tcmalloc-gperf","environment":{"distmod":"windows"}}}}
{"t":{"$date":"2026-05-15T23:06:05.157+03:00"},"s":"I", "c":"CONTROL", "id":51765, "svc":"S", "ctx":"initandlisten","msg":"Operating System","attr":{"os":{"name":"Microsoft Windows Workstation (build 26200)","version":"10.0 (build 26200)}}}}
{"t":{"$date":"2026-05-15T23:06:05.157+03:00"},"s":"I", "c":"CONTROL", "id":21951, "svc":"S", "ctx":"initandlisten","msg":"Options set by command line","attr":{"options":{"net":{"bindIp":"localhost","port":27022},"replication":{"replSet":"shard2rs"},"sharding":{"clusterRole":"shardsvr"},"storage":{"dbPath":"D:\\data\\shard2"}}}}
{"t":{"$date":"2026-05-15T23:06:05.159+03:00"},"s":"I", "c":"NETWORK", "id":11621101, "svc":"S", "ctx":"initandlisten","msg":"Overriding max connections to honor capMemoryConsumptionForPreAuthBuffers settings","attr":{"limit":100939}}
{"t":{"$date":"2026-05-15T23:06:05.159+03:00"},"s":"I", "c":"NETWORK", "id":4648601, "svc":"S", "ctx":"initandlisten","msg":"Implicit TCP FastOpen unavailable. If TCP FastOpen is required, set at least one of the related parameters","attr":{"relatedParameters":["tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"]}}}
```

Запуск другого вузла зберігання даних (Shard Server) для розподіленої системи.

```
Windows PowerShell
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PS C:\Users\anast> mongosh --port 27022
Current Mongosh Log ID: 6a077cc9ee99c2ac86abc113
Connecting to: mongodb://127.0.0.1:27022/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB: 8.2.7
Using Mongosh: 2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-05-15T23:06:05.256+03:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> rs.initiate({
  _id: 'shard2rs',
  members: [
    { _id: 0, host: 'localhost:27022' }
  ]
})
ok: 1,
{'$clusterTime': {
  clusterTime: Timestamp({ t: 1778875614, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
    keyId: Long(0)
  },
  operationTime: Timestamp({ t: 1778875614, i: 1 })
},
shard2rs [direct: secondary] test> |
```

Ініціалізація набору реплік для другого шарду кластера.



Запуск маршрутизатора mongos для координації шардованого кластера.

```

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\lanast> mongosh --port 27020
Current Mongosh Log ID: 6a077d01806908bae5abe113
Connecting to:   mongodb://127.0.0.1:27020/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB:  8.2.7
Using Mongosh:  2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-05-15T23:07:16.933+03:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

[direct: mongos] test> sh.addShard("shard1rs/localhost:27021")
{
  shardAdded: 'shard1rs',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778875667, i: 57 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778875667, i: 57 })
}
[direct: mongos] test> sh.addShard("shard2rs/localhost:27022")
{
  shardAdded: 'shard2rs',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778875680, i: 55 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778875680, i: 55 })
}

```

Рис. 7.X — Реєстрація шардованих вузлів у маршрутизаторі кластера.

```

    operationTime: Timestamp({ t: 1778875680, i: 55 })
  }
}
[direct: mongos] test> sh.enableSharding("performance_test")
{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778875693, i: 21 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778875693, i: 21 })
}
[direct: mongos] test> sh.shardCollection("performance_test.sales", { category: 1 })
{
  collectionsSharded: 'performance_test.sales',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1778875709, i: 35 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1778875709, i: 35 })
}
[direct: mongos] test> sh.status()
shardingVersion
{
  _id: ObjectId('6a0775f1f8479524f6c6ff6f'),
  clusterId: ObjectId('6a0775f1f8479524f6c6ff6e')
}
---
shards
[
  {
    _id: 'shard1rs',
    host: 'shard1rs/localhost:27021',
    state: 1,
    topologyTime: Timestamp({ t: 1778875667, i: 32 }),

```

Активація шардінгу для бази даних та налаштування розподілу колекції.

```

},
{
  _id: 'shard2rs',
  host: 'shard2rs/localhost:27022',
  state: 1,
  topologyTime: Timestamp({ t: 1778875680, i: 32 }),
  replSetConfigVersion: Long('-1')
}
]
---
active mongoses
[ { '8.2.7': 1 } ]
---
autosplit
{ 'Currently enabled': 'yes' }
---
balancer
{
  'Currently enabled': 'yes',
  'Currently running': 'no',
  'Failed balancer rounds in last 5 attempts': 0,
  'Migration Results for the last 24 hours': 'No recent migrations'
}
---
shardedDataDistribution
[
  {
    ns: 'performance_test.sales',
    shards: [
      {
        shardName: 'shard2rs',
        numOrphanedDocs: 0,
        numOwnedDocuments: 0,
        ownedSizeBytes: 0,
        orphanedSizeBytes: 0
      }
    ]
  }
]
---
databases

```

Моніторинг системних процесів балансування та розподілу даних у кластері.

```

[
  {
    database: { _id: 'config', primary: 'config', partitioned: true },
    collections: {}
  },
  {
    database: {
      _id: 'performance_test',
      primary: 'shard2rs',
      version: {
        uuid: UUID('1ca5fcd4-a05a-469e-ac13-727d7bd978f1'),
        timestamp: Timestamp({ t: 1778875693, i: 9 }),
        lastMod: 1
      }
    },
    collections: {
      'performance_test.sales': {
        shardKey: { category: 1 },
        unique: false,
        balancing: true,
        allowMigrations: true,
        chunkMetadata: [ { shard: 'shard2rs', nChunks: 1 } ],
        chunks: [
          { min: { category: MinKey() }, max: { category: MaxKey() }, 'on shard': 'shard2rs', 'last modified': Timestamp({ t: 1, i: 0 }) }
        ],
        tags: []
      }
    }
  }
]
[direct: mongos] test> |

```

Перевірка метаданих та топології шардованої колекції.

Висновок: У ході виконання лабораторної роботи було успішно реалізовано механізми реплікації та шардінгу для забезпечення високої доступності та масштабованості бази даних MongoDB. На практиці опановано процеси розгортання Replica Set для автоматичного копіювання даних та створення

розподіленого кластера через Config Server і маршрутизатор mongos. Аналіз результатів підтвердив ефективність використання ключів шардування для рівномірного балансування навантаження між різними вузлами системи.